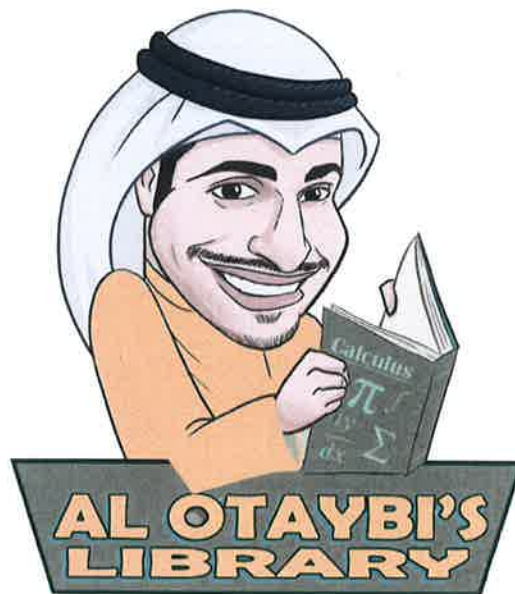


Kuwait University

2021/2022

Fall course



Calculus B

First

Section : **7.4 (Part b/c)**

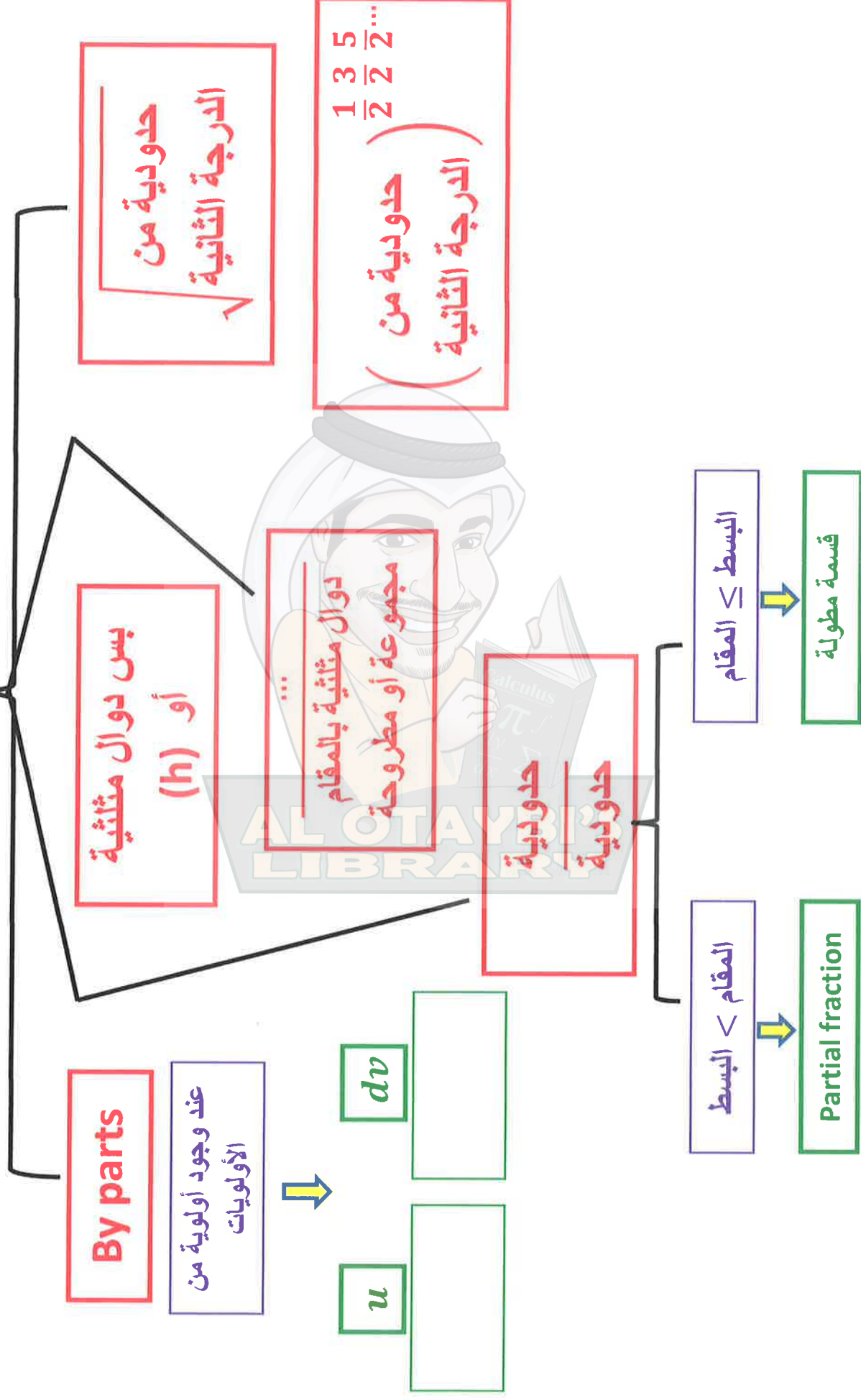
7.4 Integration of Rational Functions by Partial Fractions.

(6 b)

Eng. Mohammad Alotaybi 90048657 Twitter: @M_Alotaybi



أنواع التكاملات





Integration of rational functions involving sin & cos



طريقة تستخدم للتعامل مع التكاملات التي تحتوي على

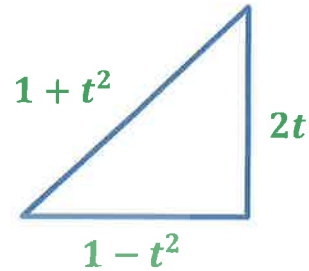
((دوال مثلثية بالمقام مجموعة أو مطروحة))

عن طريق التعويض بالاتي

$$t = \tan\left(\frac{x}{2}\right)$$

$$\sin x = \frac{2t}{1+t^2}, \quad \cos x = \frac{1-t^2}{1+t^2}$$

$$dx = \frac{2}{1+t^2} dt$$



شروط الحل

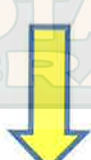
عند وجود sin - cos
يجب أن تكون زواياهم
 x

عند وجود tan
يجب أن تكون زاويتها
 $\frac{x}{2}$

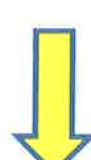
التعامل مع
sin - cos - tan
فقط



If not



If not



يتم تحويل الزاوية عن طريق

(1) القوانين

or

u

(2) فرض الزاوية

يتم تحويل tan

note that

$$\tan x = \frac{\sin x}{\cos x}$$

يتم التحويل اليهم

note that

$$\csc x = \frac{1}{\sin x}, \quad \sec x = \frac{1}{\cos x}$$

$$\cot x = \frac{1}{\tan x}$$

$$\int \frac{1}{2 + \sin x + \cos x} dx$$

دوال مثلثية بالمقام
مجموعة أرفطروعة

Let $t = \tan\left(\frac{x}{2}\right)$

$$\int \frac{1}{2 + \frac{2t}{1+t^2} + \frac{1-t^2}{1+t^2}} \cdot \frac{2}{1+t^2} dt$$

$$2 \int \frac{1}{2(1+t^2) + 2t + 1 - t^2} dt$$

$$2 + 2t^2 + 2t + 1 - t^2$$

$$t^2 + 2t + 3$$

$$2 \int \frac{1}{t^2 + 2t + 3} dt$$

$$t^2 + 2t + 3$$

$$(t + 1)^2 - 1 + 3$$

$$(t + 1)^2 + 2$$

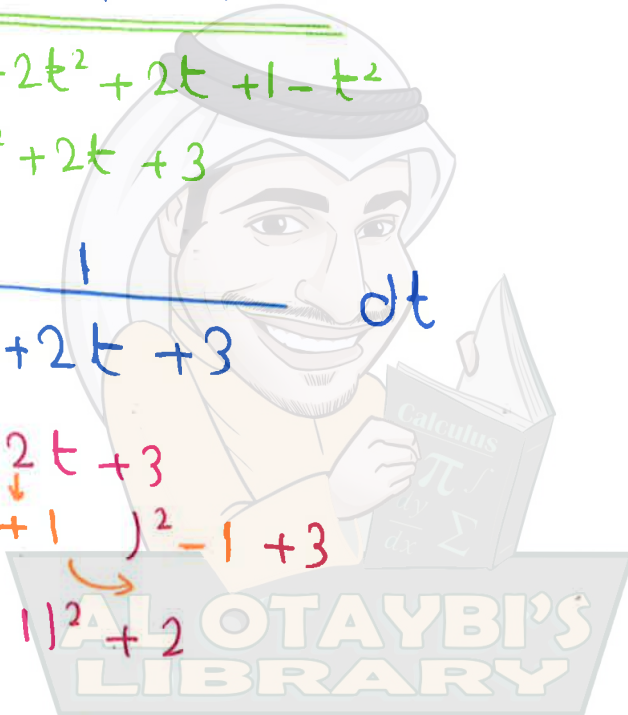
$$2 \int \frac{1}{(t+1)^2 + 2} dt$$

$$2 \cdot \frac{1}{\sqrt{2}} \tan^{-1}\left(\frac{t+1}{\sqrt{2}}\right) + C$$

$$\frac{2}{\sqrt{2}} \tan^{-1}\left(\frac{\tan \frac{x}{2} + 1}{\sqrt{2}}\right) + C$$

* الشروط *

- ✓ (1)
- ✓ (2)
- ✓ (3)



$$\int \frac{\cos x}{\sin x + \cos x + 1} dx$$

دوال مثلثية بالمقام
مجموعه أرقطرية

* الشروط *

✓	(1)
✓	(2)
✓	(3)

Let $t = \tan\left(\frac{x}{2}\right)$

$$\int \frac{\frac{1-t^2}{1+t^2}}{\frac{2t}{1+t^2} + \frac{1-t^2}{1+t^2} + 1} \cdot \frac{2}{1+t^2} dt$$

$$2 \int \frac{\frac{1-t^2}{1+t^2}}{2t + 1 - t^2 + 1 + t^2} dt = 2 \int \frac{\frac{1-t^2}{1+t^2}}{(2t+2)} dt$$

$$2 \int \frac{1-t^2}{(2t+2)(1+t^2)} dt = 2 \int \frac{(1-t)(1+t)}{2(t+1)(1+t^2)} dt$$

$$\frac{2}{2} \int \frac{1-t}{1+t^2} dt$$

$$\int \frac{-\frac{1}{2}(2t) + 1}{1+t^2} dt$$

$$\int \left[\frac{-\frac{1}{2}(2t)}{1+t^2} + \frac{1}{1+t^2} \right] dt$$

$$= -\frac{1}{2} \ln|1+t^2| + \tan^{-1}t + c$$

$$= -\frac{1}{2} \ln|1+(\tan\frac{x}{2})^2| + \tan^{-1}\tan\frac{x}{2} + c$$

$$= -\frac{1}{2} \ln|1+(\tan\frac{x}{2})^2| + \frac{x}{2} + c$$



Text Book Question



قراءة

$$\int \frac{1}{1 + \sin x - \cos x} dx$$

Let $t = \tan\left(\frac{x}{2}\right)$

$$\int \frac{1}{1 + \frac{2t}{1+t^2} - \frac{1-t^2}{1+t^2}} \cdot \frac{2}{1+t^2} dt$$

$$2 \int \frac{1}{1+t^2+2t-(1-t^2)} dt$$

$$\begin{aligned} 1+t^2+2t-1+t^2 \\ 2t^2+2t \\ 2t(t+1) \end{aligned}$$

$$2 \int \frac{1}{2t(t+1)} dt$$

$$\frac{1}{t(t+1)} = \frac{A}{t} + \frac{B}{t+1}$$

* $t(t+1)$

$$1 = A(t+1) + B \cdot t$$

$t=0$ $1 = A(1) + 0 \rightarrow A=1$

$t=-1$ $1 = 0 + B(-1) \rightarrow B=-1$

$$\begin{aligned} \int \left[\frac{1}{t} + \frac{-1}{t+1} \right] dt &= \ln|t| - \ln|t+1| + c \\ &= \ln\left|\tan\left(\frac{x}{2}\right)\right| - \ln\left|\tan\left(\frac{x}{2}\right)+1\right| + c \end{aligned}$$



Text Book Question



قراءة

$$\int \frac{1}{3\sin x - 4\cos x} dx$$

Let $t = \tan\left(\frac{x}{2}\right)$

$$\int \frac{1}{3 \frac{2t}{1+t^2} - 4 \frac{1-t^2}{1+t^2}} \cdot \frac{2}{1+t^2} dt$$

$$2 \int \frac{1}{6t - 4(1-t^2)} dt$$

$$6t - 4 + 4t^2$$

$$2(2t^2 + 3t - 2)$$

$$\cancel{2} \int \frac{1}{\cancel{2}(2t^2 + 3t - 2)} dt$$

$$\frac{1}{(2t-1)(t+2)} = \frac{A}{2t-1} + \frac{B}{t+2}$$

$$*(2t-1)(t+2)$$

$$1 = A(t+2) + B(2t-1)$$

$$\underline{t = -2} \quad 1 = 0 + B(-5) \longrightarrow B = \frac{-1}{5}$$

$$\underline{t = \frac{1}{2}} \quad 1 = A\left(\frac{1}{2} + 2\right) + 0 \longrightarrow 1 = A\left(\frac{5}{2}\right) \longrightarrow A = \frac{2}{5}$$

$$\int \left[\frac{\frac{2}{5}}{2t-1} - \frac{\frac{-1}{5}}{t+2} \right] dt = \frac{2}{5} \int \frac{1}{2t-1} dt - \frac{-1}{5} \int \frac{1}{t+2} dt$$

$$= \frac{2}{5} \cdot \frac{1}{2} \ln|2t-1| + \frac{1}{5} \ln|t+2| + c = \frac{1}{5} \ln|2\tan\left(\frac{x}{2}\right) - 1| + \frac{1}{5} \ln|\tan\frac{x}{2} + 2| + c$$

$$\int \frac{2 + \tan\left(\frac{x}{2}\right)}{2\sin x + 2\cos x + 3} dx$$

Let $t = \tan\left(\frac{x}{2}\right)$

$$\int \frac{2 + t}{2 \frac{2t}{1+t^2} + 2 \frac{1-t^2}{1+t^2} + 3} dt$$

$$2 \int \frac{2 + t}{2(2t) + 2(1-t^2) + 3(1+t^2)} dt$$

$$4t + 2 - 2t^2 + 3 + 3t^2$$

$$t^2 + 4t + 5$$

$$2 \int \frac{2 + t}{t^2 + 4t + 5} dt$$

$$2 \int \frac{\frac{1}{2} (2t + 4)}{t^2 + 4t + 5} dt$$

$$2 \cdot \frac{1}{2} \ln |t^2 + 4t + 5| + C$$

$$\ln |(\tan \frac{x}{2})^2 + 4(\tan \frac{x}{2}) + 5| + C$$

$$\int_0^{\frac{\pi}{2}} \frac{dx}{1 + \sin x + \cos x}$$

Fall 2021

Let $t = \tan\left(\frac{x}{2}\right)$

$$\int \frac{1}{1 + \frac{2t}{1+t^2} + \frac{1-t^2}{1+t^2}} \cdot \frac{2}{1+t^2} dt$$

$$2 \int \frac{1}{1+t^2 + 2t + 1-t^2} dt$$

$$2 \int \frac{2t+2}{2(t+1)} dt$$

$$= \ln|t+1|$$

$$= \left[\ln \left| \tan\left(\frac{x}{2}\right) + 1 \right| \right]_0^{\frac{\pi}{2}}$$

$$= \ln \left| \tan\left(\frac{\pi}{4}\right) + 1 \right| - \ln |0+1|$$

$$= \ln |1+1| - 0$$

$$= \ln 2$$

$$\int \frac{dx}{2\sin x - \cos x}$$

Nov 2016

Let $t = \tan\left(\frac{x}{2}\right)$

$$\int \frac{1}{2 \frac{2t}{1+t^2} - \frac{1-t^2}{1+t^2}} \cdot \frac{2}{1+t^2} dt$$

$$\int \frac{2}{4t - (1-t^2)} dt$$

$$\frac{4t - 1 + t^2}{4t - 1 + t^2}$$

$$\int \frac{2}{t^2 + 4t - 1} dt$$

$$\frac{t^2 + 4t - 1}{(t+2)^2 - 4 - 1}$$

$$\int \frac{2}{(t+2)^2 - 5} dt$$

$$2 \cdot \frac{1}{2\sqrt{5}} \ln \left| \frac{t+2-\sqrt{5}}{t+2+\sqrt{5}} \right| + C$$

$$\frac{1}{\sqrt{5}} \ln \left| \frac{\tan\left(\frac{x}{2}\right) + 2 - \sqrt{5}}{\tan\left(\frac{x}{2}\right) + 2 + \sqrt{5}} \right| + C$$

$$\int \frac{dx}{\left(\sin \frac{x}{2} + \cos \frac{x}{2}\right)^2}$$



دائما قبل وتطبيق الشروط
عند امكانية التبسيط عن طريق القانون التالي فانه يتم ذلك

$$\cos^2 x + \sin^2 x = 1$$



$$\sin^2 \frac{x}{2} + 2 \sin \frac{x}{2} \cos \frac{x}{2} + \cos^2 \frac{x}{2}$$

$$\sin^2 \frac{x}{2} + \cos^2 \frac{x}{2} + 2 \sin \frac{x}{2} \cos \frac{x}{2}$$

$$1 + 2 \sin \frac{x}{2} \cos \frac{x}{2}$$

$$\int \frac{dx}{1 + 2 \sin \frac{x}{2} \cos \frac{x}{2}}$$

$$\sin 2x = 2 \sin x \cos x$$

$$\Rightarrow \sin x = 2 \sin \frac{x}{2} \cos \frac{x}{2}$$

$$\int \frac{dx}{1 + \sin x}$$

sol #1

sol #2

$$\int \frac{dx}{1 + \sin x}$$

$$\int \frac{1 - \sin x}{1 - \sin^2 x} dx$$

$$\cos^2 x + \sin^2 x = 1$$

$$\cos^2 x = 1 - \sin^2 x$$

$$\int \frac{1 - \sin x}{\cos^2 x} dx$$

$$\int \left(\frac{1}{\cos^2 x} - \frac{\sin x}{\cos^2 x} \right) dx$$

$$\int \sec^2 x dx - \int \sin x (\cos x)^{-2} dx$$

$\tan x$

$$u = \cos x$$

$$du = -\sin x dx$$

$$-du = \sin x dx$$

$$+ \int u^{-2} du$$

$$\frac{u^{-1}}{-1} = -\frac{1}{u} + C$$

$$= \frac{1}{\cos x} + C$$

دوال مثلثية بالمقام
مجموعة أو مطروقة

* الشروط *

✓ (1)

✓ (2)

X (3)

See Next page
plz!

$$\int \frac{1}{1 + \sin x} dx \rightarrow$$

أضرب على جنب

Let $t = \tan\left(\frac{x}{2}\right)$

$$\int \frac{1}{1 + \frac{2t}{1+t^2}} \cdot \frac{2}{1+t^2} dt$$

$$2 \int \frac{1}{1+t^2 + 2t} dt$$

$$2 \int \frac{1}{t^2 + 2t + 1} dt$$

$$(t+1)^2 - 1 + 1$$

$$(t+1)^2$$

$$2 \int \frac{1}{(t+1)^2} dt = 2 \int (t+1)^{-2} dt$$

$$= 2 \frac{(t+1)^{-1}}{-1} + C$$

$$= \frac{-2}{t+1} + C$$

$$= \frac{-2}{\tan\left(\frac{x}{2}\right) + 1} + C$$

$$\int \frac{2}{\sin^2(4x) + (1 + \cos(4x))^2} dx$$



دائما قبل وتطبيق الشروط
عند امكانية التبسيط عن طريق القانون التالي فانه يتم ذلك

$$\cos^2 x + \sin^2 x = 1$$



$$\sin^2(4x) + 1 + 2\cos(4x) + \cos^2(4x)$$

$$\sin^2(4x) + \cos^2(4x) + 1 + 2\cos(4x)$$

$$1 + 1 + 2\cos(4x)$$

$$2 + 2\cos(4x) = 2(1 + \cos(4x))$$

$$\int \frac{2}{2(1 + \cos(4x))} dx$$

$$\int \frac{1}{1 + \cos(4x)} dx$$

Sol #2

$$\int \frac{1}{1 + \cos(4x)} dx$$

$$\frac{1 - \cos(4x)}{1 - \cos(4x)}$$

$$\int \frac{1 - \cos(4x)}{1 - \cos^2(4x)} dx$$

$$\rightarrow \cos^2(4x) + \sin^2(4x) = 1$$

$$\sin^2(4x) = 1 - \cos^2(4x)$$

$$\int \frac{1 - \cos(4x)}{\sin^2(4x)} dx$$

$$\int \left(\frac{1}{\sin^2(4x)} - \frac{\cos(4x)}{\sin^2(4x)} \right) dx$$

$$\int \csc^2(4x) dx - \int \frac{\cos(4x)}{\sin^2(4x)} dx$$

$$u = \sin(4x)$$

$$du = 4\cos(4x) dx$$

$$\frac{1}{4} du = \cos(4x) dx$$

$$\frac{1}{4} \int u^{-2} du$$

$$\frac{1}{4} \frac{u^{-1}}{-1} + C$$

$$-\frac{1}{4} (\sin(4x))^{-1} + C$$

Sol #1

$$\text{Let } u = 4x$$

$$du = 4 dx$$

$$\frac{1}{4} du = dx$$

$$\frac{1}{4} \int \frac{1}{1 + \cos u} du$$

See Next page
plz!

- * الشروط *
- ✓ (1)
 - ✓ (2)
 - ✗ (3)

$$\frac{1}{4} \int \frac{1}{1 + \cos u} du$$

Let $t = \tan\left(\frac{u}{2}\right)$

$$\frac{1}{4} \int \frac{1}{1 + \frac{1-t^2}{1+t^2}} \cdot \frac{2}{1+t^2} dt$$

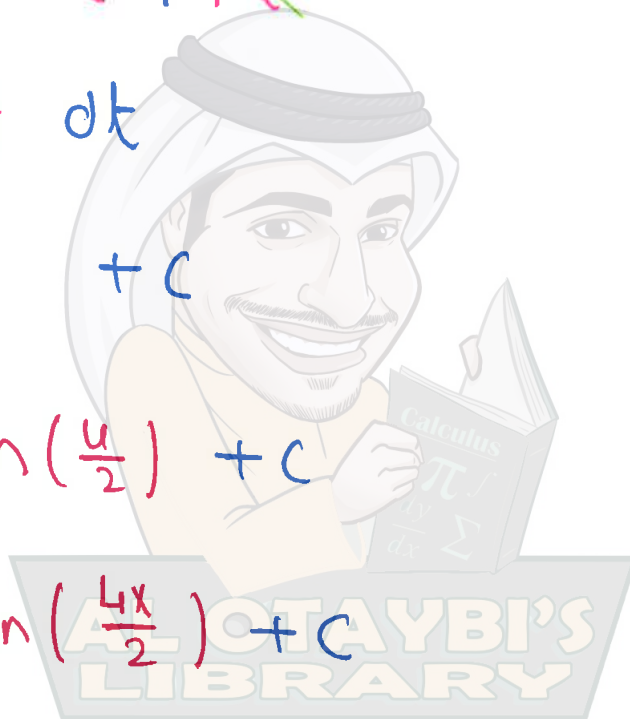
$$\frac{2}{4} \int \frac{1}{\cancel{1+t^2} + \cancel{1-t^2}} dt$$

$$\frac{1}{2} \int \frac{1}{2} dt$$

$$\frac{1}{4} t + C$$

$$\frac{1}{4} \tan\left(\frac{u}{2}\right) + C$$

$$\frac{1}{4} \tan\left(\frac{4x}{2}\right) + C$$



$$\int \frac{\operatorname{csch}^2 x}{1 + \sin(\operatorname{coth} x)} dx$$

لاحظ أن الزاوية ليس
حدودية من الدرجة الأولى

u

راح أسميها



Let $u = \operatorname{coth} x$

$$du = -\operatorname{csch}^2 x dx$$

$$-du = \operatorname{csch}^2 x dx$$

$$-\int \frac{1}{1 + \sin u} du$$

Sol #1

Let $t = \tan\left(\frac{u}{2}\right)$

$$-\int \frac{1}{1 + \frac{2t}{1+t^2}} dt$$

Sol #2

حل

مرفقا

AL OTAYBI'S
LIBRARY

...

$$\int \frac{\csc x}{2\csc x + 1} dx$$

دوال مثلثية بالمقام
مجموعة أو مفردة

الشروط	
X	(1)
✓	(2)
✓	(3)

$$\int \frac{\frac{1}{\sin x}}{\frac{2}{\sin x} + 1} dx$$

$$\int \frac{\frac{1}{\sin x}}{\frac{2}{\sin x} + \frac{\sin x}{\sin x}} dx = \int \frac{\frac{1}{\sin x}}{\frac{2 + \sin x}{\sin x}} dx$$

$$\int \frac{1}{2 + \sin x} dx$$

Let $t = \tan\left(\frac{x}{2}\right)$

$$\int \frac{1}{2 + \frac{2t}{1+t^2}} \cdot \frac{2}{1+t^2} dt$$

$$2 \int \frac{1}{2(1+t^2) + 2t} dt = \int \frac{1}{t^2 + t + 1} dt$$

$$\begin{aligned} & \left(t + \frac{1}{2}\right)^2 - \frac{1}{4} + 1 \\ & \left(t + \frac{1}{2}\right)^2 - \frac{1}{4} + \frac{4}{4} \\ & \left(t + \frac{1}{2}\right)^2 + \frac{3}{4} \end{aligned}$$

$$2 \int \frac{1}{2(t^2 + t + 1)} dt$$

$$\int \frac{1}{\left(t + \frac{1}{2}\right)^2 + \frac{3}{4}} dt$$

$$= \frac{1}{\sqrt{3/4}} \tan^{-1} \left(\frac{t + \frac{1}{2}}{\sqrt{3/4}} \right) + C$$

$\tan\left(\frac{x}{2}\right)$

$$\int \frac{\sec x}{3 \tan x - 4} dx$$

* السؤاا *

X (1)
X (2)
✓ (3)

$$\int \frac{\frac{1}{\cos x}}{3 \tan x - 4} dx$$

$$\int \frac{\frac{1}{\cos x}}{\frac{3 \sin x}{\cos x} - 4} dx = \int \frac{\frac{1}{\cancel{\cos x}}}{\frac{3 \sin x - 4 \cancel{\cos x}}{\cancel{\cos x}}} dx$$

$$\int \frac{1}{3 \sin x - 4 \cos x} dx$$

Let $t = \tan\left(\frac{x}{2}\right)$

$$\int \frac{1}{3 \frac{2t}{1+t^2} - 4 \frac{1-t}{1+t^2}} dt$$

$$2 \int \frac{1}{3(2t) - 4(1-t^2)} dt$$

$$6t - 4 + 4t^2$$

$$2(2t^2 + 3t - 2)$$

$$2 \int \frac{1}{2(2t^2 + 3t - 2)} dt$$

$$\int \frac{1}{2t^2 + 3t - 2} dt$$

Sol #1

المقام يفتل

$$\frac{1}{(2t-1)(t+2)} = \frac{A}{2t-1} + \frac{B}{t+2}$$

* $(2t-1)(t+2)$

$$1 = A(t+2) + B(2t-1)$$

$t = -2$

$$1 = 0 + B(-4-1) \rightarrow B = -\frac{1}{5}$$

$t = \frac{1}{2}$

$$1 = A\left(\frac{1}{2} + 2\right) + 0$$

$$1 = A\left(\frac{5}{2}\right) \rightarrow A = \frac{2}{5}$$

$$\int \left[\frac{\frac{2}{5}}{2t-1} + \frac{\frac{-1}{5}}{t+2} \right] dx$$

$$\frac{2}{5} \int \frac{1}{2t-1} dt - \frac{1}{5} \int \frac{1}{t+2} dt$$

$$\frac{2}{5} \cdot \frac{1}{2} \ln|2t-1| - \frac{1}{5} \ln|t+2| + C$$

$$\frac{1}{5} \ln|2 \tan\left(\frac{x}{2}\right) - 1| - \frac{1}{5} \ln|\tan\frac{x}{2} + 1| + C$$

Sol #2

$$2t^2 + 3t - 2$$

$$2 \left[t^2 + \frac{3}{2}t - 1 \right]$$

$$2 \left[\left(t + \frac{3}{4} \right)^2 - \frac{9}{16} - 1 \right]$$

$$2 \left[\left(t + \frac{3}{4} \right)^2 - \frac{25}{16} \right]$$

$$\int \frac{1}{2 \left[\left(t + \frac{3}{4} \right)^2 - \frac{25}{16} \right]} dt$$

$$\frac{1}{2} \cdot \frac{1}{2 \left(\frac{5}{4} \right)} \ln \left| \frac{t + \frac{3}{4} - \frac{5}{4}}{t + \frac{3}{4} + \frac{5}{4}} \right| + C$$

$\tan\left(\frac{x}{2}\right)$

$$\int \frac{\sec x}{2\sec x + \tan x + 1} dx$$

* الشروط *

x	(1)
x	(2)
	(3)

$$\int \frac{\frac{1}{\cos x}}{\frac{2}{\cos x} + \tan x + 1} dx$$

$$\int \frac{\frac{1}{\cos x}}{\frac{2}{\cos x} + \frac{\sin x}{\cos x} + 1} dx = \int \frac{\frac{1}{\cos x}}{\frac{2}{\cos x} + \frac{\sin x}{\cos x} + \frac{\cos x}{\cos x}} dx$$

$$\int \frac{\frac{1}{\cancel{\cos x}}}{\frac{2 + \sin x + \cancel{\cos x}}{\cancel{\cos x}}} dx = \int \frac{1}{2 + \sin x + \cos x} dx$$

Let $t = \tan \frac{x}{2}$

$$\int \frac{1}{2 + \frac{2t}{1+t^2} + \frac{1-t^2}{1+t^2}} dx$$

$$2 \int \frac{1}{\frac{2(1+t^2) + 2t + 1 - t^2}{1+t^2}} dx = 2 \int \frac{1}{(t+1)^2 + 2} dt$$

$2 + 2t^2 + 2t + 1 - t^2$
 $t^2 + 2t + 3$

$$2 \int \frac{1}{t^2 + 2t + 3} dx$$

$(t + 1)^2 - 1 + 3$
 $(t + 1)^2 + 2$

$$= 2 \frac{1}{\sqrt{2}} \tan^{-1} \left(\frac{t+1}{\sqrt{2}} \right) + C$$

$$= \frac{2}{\sqrt{2}} \tan^{-1} \left(\frac{\tan(\frac{x}{2}) + 1}{\sqrt{2}} \right) + C$$

$$\int \frac{\csc x}{(1 + \cos x)^2} dx$$

* الشرط 1 *

X (1)
✓ (2)
✓ (3)

$$\int \frac{\frac{1}{\sin x}}{(1 + \cos x)^2} dx$$

$$\int \frac{1}{\sin x (1 + \cos x)^2} dx$$

Let $t = \tan \frac{x}{2}$

$$\int \frac{1}{\frac{2t}{1+t^2} \left(1 + \frac{1-t^2}{1+t^2}\right)^2} \cdot \frac{2}{1+t^2} dt$$

توحيد المقامات

$$\int \frac{1}{2t \left(\frac{1+t^2}{1+t^2} + \frac{1-t^2}{1+t^2}\right)^2} dt = \int \frac{1}{t \left(\frac{1+t^2+1-t^2}{1+t^2}\right)^2} dt$$

$$\int \frac{1}{t \left(\frac{2}{1+t^2}\right)^2} dt = \int \frac{1}{t \frac{4}{(1+t^2)^2}} dt$$

$$\int \frac{(1+t^2)^2}{4t} dt = \frac{1}{4} \int \frac{1+2t^2+t^4}{t} dt$$

$$\frac{1}{4} \int \left[\frac{1}{t} + \frac{2t^2}{t} + \frac{t^4}{t} \right] dt$$

$$\frac{1}{4} \left[\ln|t| + 2 \frac{t^2}{2} + \frac{t^4}{4} \right] + C$$

$$\frac{1}{4} \left[\ln|\tan \frac{x}{2}| + (\tan \frac{x}{2})^2 + \frac{(\tan \frac{x}{2})^4}{4} \right] + C$$

$$\int \frac{dx}{1 + 2\sec x}$$

* الشروط *

x	(1)
✓	(2)
✓	(3)

$$\int \frac{dx}{1 + \frac{2}{\cos x}} = \int \frac{1}{\frac{\cos x}{\cos x} + \frac{2}{\cos x}} dx$$

$$\int \frac{1}{\frac{\cos x + 2}{\cos x}} dx = \int \frac{\cos x}{\cos x + 2} dx$$

Let $t = \tan \frac{x}{2}$

$$\int \frac{\frac{1-t^2}{1+t^2}}{\frac{1-t^2}{1+t^2} + 2} \cdot \frac{2}{1+t^2} dt$$

$$2 \int \frac{\frac{1-t^2}{1+t^2}}{1-t^2 + 2(1+t^2)} dt$$

$$\frac{1-t^2 + 2 + 2t^2}{t^2 + 3}$$

$$2 \int \frac{\frac{1-t^2}{1+t^2}}{t^2 + 3} dt = 2 \int \frac{1-t^2}{(t^2+3)(1+t^2)} dt$$

$$\frac{1-t^2}{(t^2+3)(t^2+1)} = \frac{At+B}{t^2+3} + \frac{Ct+D}{t^2+1}$$

...

$$\int \frac{dx}{\tan x + \sin x}$$

* الشروط *

- ✓ (1)
- x (2)
- (3)

$$\int \frac{1}{\frac{\sin x}{\cos x} + \sin x} dx$$

$$\int \frac{1}{\frac{\sin x}{\cos x} + \frac{\sin x \cos x}{\cos x}} dx = \int \frac{1}{\frac{\sin x + \sin x \cos x}{\cos x}} dx$$

$$\int \frac{\cos x}{\sin x + \sin x \cos x} dx$$

يُفضل التبسيط
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$$\int \frac{\cos x}{\sin x (1 + \cos x)} dx$$

Let $t = \tan \frac{x}{2}$

$$\int \frac{\frac{1-t^2}{1+t^2}}{\frac{2t}{1+t^2} \left(1 + \frac{1-t^2}{1+t^2}\right)} \cdot \frac{2}{1+t^2} dt = 2 \int \frac{\frac{1-t^2}{1+t^2}}{2t \left(\frac{1+t^2+1-t^2}{1+t^2}\right)} dt$$

$$2 \int \frac{1-t^2}{2t(2)} dt = \frac{1}{2} \int \frac{1-t^2}{t} dt$$

$$\frac{1}{2} \int \left[\frac{1}{t} - \frac{t^2}{t} \right] dt$$

$$\frac{1}{2} \left[\ln|t| - \frac{t^2}{2} \right] + C$$

$$\frac{1}{2} \left[\ln \left| \tan \frac{x}{2} \right| - \frac{(\tan \frac{x}{2})^2}{2} \right] + C$$

$$\int \frac{\tan\left(\frac{5x}{2}\right)}{1 - \sin 5x + \cos 5x} dx$$

* الشروط *

- ✓ (1)
- ✗ (2)
- ✗ (3)

Let $u = 5x$
 $du = 5 dx \rightarrow \frac{1}{5} du = dx$

$$\frac{1}{5} \int \frac{\tan\left(\frac{u}{2}\right)}{1 - \sin u + \cos u} du$$

Let $t = \tan\left(\frac{u}{2}\right)$

$$\frac{1}{5} \int \frac{t}{1 - \frac{2t}{1+t^2} + \frac{1-t^2}{1+t^2}} \cdot \frac{2}{1+t^2} dt$$

$$\frac{2}{5} \int \frac{t}{1+t^2 - 2t + 1-t^2} dt$$

$$2 - 2t = 2(1-t)$$

$$\frac{2}{5} \int \frac{t}{2(1-t)} dt \Rightarrow \frac{1-t}{1-t} \sqrt{\frac{-1}{t} \oplus \frac{1}{-1}} = \frac{1}{1-t}$$

$$\frac{1}{5} \int \left[-1 + \frac{1}{1-t} \right] dt$$

$$\frac{1}{5} \left[-t - \ln|1-t| \right] + c$$

$$\frac{1}{5} \left[-\tan\left(\frac{u}{2}\right) - \ln|1 - \tan\left(\frac{u}{2}\right)| \right] + c$$

$\xrightarrow{5x}$



Multiple choice



Using $t = \tan\left(\frac{x}{2}\right)$ gives $\int \frac{1}{\sin x - \cos x} dx =$

(a) $\int \frac{1}{2t^2 + 2t - 1} dt$

(b) $\int \frac{1}{t^2 - 2t - 1} dt$

(c) $\int \frac{2}{(t+1)^2} dt$

(d) $\int \frac{2}{t^2 - 2t + 1} dt$

(e) $\int \frac{2}{t^2 + 2t - 1} dt$

Oct 2017

Let $t = \tan\left(\frac{x}{2}\right)$

$$\int \frac{1}{\frac{2t}{1+t^2} - \frac{1-t^2}{1+t^2}} dt$$

$$\int \frac{2}{2t - (1-t^2)} dt$$

$$\int \frac{2}{t^2 + 2t - 1} dt$$

(e)